

Question:
What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

- Answer Choices:
- A. 2.48×10^{-8} cm
 - B. 5.92×10^7 cm/sec
 - C. 8.66×10^{-6} cm
 - D. 3.14×10^{-5} cm
 - E. 9.81×10^{-4} cm
 - F. 5.27×10^{-9} m
 - G. 1.23×10^{-7} cm
 - H. 4.14×10^{-10} m
 - I. 6.625×10^{-27} erg sec
 - J. 0.512×10^6 eV

Reasoning + Answer:

<think>
kinetic energy in electron volts (eV) to joules (J): $1 \text{ eV} = 1.602 \times 10^{-16} \text{ J}$, so $1 \text{ eV} = 1.602 \times 10^{-16} \text{ J}$
1 eV in electron volts:
 $1 \text{ eV} = 1.602 \times 10^{-16} \text{ J}$
1 eV in joules
 $1 \text{ eV} = 1.602 \times 10^{-16} \text{ J}$
de Broglie wavelength (λ) formula: $\lambda = h/p$
where $h = 6.625 \times 10^{-34} \text{ J}\cdot\text{s}$ and $p = (2mE)^{1/2}$ in terms of momentum (p)
1) Use KE to find the momentum:
 $KE = (1/2)mv^2$; $v^2 = 2KE/m$
 $v = \sqrt{2KE/m}$
2) Substitute KE and m (electron mass $9.109 \times 10^{-31} \text{ kg}$) into the momentum formula
 $p = (2m(1.602 \times 10^{-16} \text{ J}))^{1/2} = (9.109 \times 10^{-31} \text{ kg})^{1/2} = 3.613 \times 10^{-16} \text{ m/s}$
3) Calculate λ :
 $\lambda = (6.625 \times 10^{-34} \text{ J}\cdot\text{s}) / (3.613 \times 10^{-16} \text{ m/s}) = 1.84 \times 10^{-18} \text{ meters}$ (this is not in the option list, so convert to nanometers and check the options)
 $1.84 \times 10^{-18} \text{ J}\cdot\text{s} = 1.84 \times 10^{-18} \times 10^9 \text{ nm} = 1.84 \times 10^{-9} \text{ nm}$
 $1.84 \times 10^{-9} \text{ nm}$ is the same as $1.84 \times 10^{-7} \text{ cm}$
4) Now check the option which matches $1.84 \times 10^{-7} \text{ cm}$
Option A is $2.48 \times 10^{-8} \text{ cm}$, which is not the same
Option C is $8.66 \times 10^{-6} \text{ cm}$, which is also not the same
Option G is $5.27 \times 10^{-9} \text{ m}$, which is $5.27 \times 10^{-9} \text{ m} = 5.27 \times 10^{-10} \text{ nm} = 1.84 \times 10^{-7} \text{ cm}$
</think>
<answer> G. $5.27 \times 10^{-9} \text{ m}$ </answer>